[CS304] Introduction to Cryptography and Network Security

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Lagrange's TheoremL:

Theorem: If G is a finite group and H is a subgroup of G, then $|H|$ divides $|G|$.

Proof:

Let $s = \{e, a, a^2, a^3, \ldots, a^{o(a)-1}\} = \langle a \rangle$.

$(s, k) \mapsto$ Subgroup of (G, K)

$|S| \mid |G|$

$o(a) \mid |G|$

If the order of a belonging to G is t, then the order of a^k will be $\frac{t}{\gcd(t,k)}$.

$\langle a \rangle = \{e, a, a^2, a^3, \ldots, a^{o(a)-1}\}$

$\langle a^t \rangle = \{e, a, a^2, a^3, \ldots, a^{(o(a)-1)^t}\}$

Let $b = a^t$.

$\langle b \rangle = \langle a^t \rangle = \{e, b, b^2, b^3, \ldots, a^{(o(b)-1)}\} = \langle a^t \rangle$

Ring

A ring $(R, +_R, \times_R)$ consists of a set R with two binary operations arbitrarily denoted by $+_R$ and $\times_R$ on R satisfying the following properties:

1. $(R, +_R)$ is an abelian group with the identity element O_R.
2. The operation $\times_R$ is associative, i.e., $a \times_R (b \times_R c) = (a \times_R b) \times_R c$.
3. There is a multiplicative identity denoted by 1_R with $1_R \neq 0_R$, i.e., $1_R \times_R a = a \times_R 1_R = a$, for all $a \in R$.
4. The operation $\times_R$ is distributive over $+_R$, i.e., $(b +_R c) \times_R a = (b \times_R a) +_R (c \times_R a)$.

Example: For $(\mathbb{Z}, +, \cdot)$, it does not form a ring.

Note: Commutative Ring

A ring $(R, +_R, \times_R)$ where $a \times_R b = b \times_R a$ for all $a, b \in R$ is called a commutative ring.

Additional Points:

1. An element a of a ring R is called a unit or invertible if there exists $b \in R$ such that $a \times_R b = 1_R$.
2. The set of units in a ring R forms a group under the multiplication operation. This group is known as the group of units of R.

Field

A field is a non-empty set F together with two binary operations $+$ and $*$ for which the following properties are satisfied:

1. $(F, +)$ is an abelian group.
2. If O_F denotes the additive identity element of $(F, +)$, then $(F \backslash \{O_F\}, *)$ is a commutative or abelian group.
3. For all $a, b, c \in F$, we have $(a * b) + c = (a * c) + (b * c)$.

Examples:

- $(\mathbb{Z}, +, \cdot)$ is not a field.
- $(\mathbb{Q}, +, \cdot)$ is a field.

Example 3: Let $F_p = \{0, 1, 2, 3, \ldots, p-1\}$ be the set of prime numbers. Then $(F_p, +_p, \times_p)$ is a field where:

- $+_p$: $(x + y) \mod p =$ commutative group.
- $\times_p$: $(x * y) \mod p =$ commutative group.

Field Extension

Suppose k_2 is a field with addition $+$ and multiplication $*$. Suppose $k_1 \subseteq k_2$ is closed under both these operations such that k_1 itself is a field for the restrictions of $+$ and $*$ to the set k_1. Then k_1 is called a subfield of k_2, and k_2 is called a field extension of k_1.

Definition: Let F be a field $(F, +, *)$. Then $F[x] = \{a_0 + a_1 x + \ldots | a_i \in F\}$ is a polynomial ring.

Addition:

$$\begin{aligned}
&(a_0 + a_1 x + \ldots + a_{n-1} x^{n-1}) \\
&\quad + (b_0 + b_1 x + \ldots + b_{n-1} x^{n-1}) \\
&\quad\quad = (a_0 + b_0) + (a_1 + b_1) x + \ldots + (a_{n-1} + b_{n-1}) x^{n-1}
\end{aligned}$$

Multiplication:

$$\begin{aligned}
&(a_0 + a_1 x + \ldots + a_{n-1} x^{n-1}) \\
&\quad * (b_0 + b_1 x + \ldots + b_{n-1} x^{n-1}) \\
&\quad\quad = (a_0 b_0) + (a_0 b_1 + a_1 b_0) x + \ldots
\end{aligned}$$

Since field multiplication and field addition are involved, $F[x]$ forms a ring because it has an identity element, additive operation, and is distributive, etc.

Definition: A polynomial $p(x)$ belongs to $F[x]$ of degree $n \geq 1$ is called irreducible if it can't be written in the form $p_1(x) \cdot p_2(x)$ with $p_1(x), p_2(x) \in F[x]$ and degree of $p_1(x), p_2(x)$ must be ≥ 1.

$I = \langle P(x) \rangle = \{q(x), p(x) \,|\, q(x) \text{ belongs to } F[x]\}$

$(F[x], +, *) \rightarrow$ ring

$F[x]$ - polynomial ring

$F[x] \mid \langle p(x) \rangle$ - irreducible polynomial

Degree of $F[x] < P[x]$

$q(x)$ belongs to $F(x)$

$q(x) = d(x) \cdot p(x) + r(x)$

$r(x)$ belongs to $F[x] \mid \langle p(x) \rangle$

$(F[x]/\langle P[x], +_{\text{mod } p(n)}, *_{\text{mod } p(n)} \rangle)$

Example: $(x^2 + 1)$ in $R[x]$ is an irreducible polynomial.